Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

So you're thinking about data science

The short version — written for you, not about you.

Everything that matters, none of the bits written for your parents.

This is the one where we're least sure of our own answer, and you should know that before anything else.

We scored twenty-seven careers on AI exposure. Entry-level data analysis came out at **8.2 out of 10** — one of the highest.

And the Bureau of Labor Statistics projects this occupation growing 33.5% by 2034 — the fourth-fastest-growing job in the entire US economy.

Those don't obviously fit together. Working out why is the useful part.

The 60-second version

- **Entry-level analysis is highly automatable.** SQL, dashboards, cleaning, standard models.
- **But demand is growing faster than almost any occupation in America.**
- Entry salaries are *rising*, not falling — analyst starting pay is up around $20,000 in a year.
- The title is surviving while what it means changes completely.
- And the big one: **being close to AI doesn't protect you from it.**

The thing most people get wrong

Here's an intuition worth killing, because a lot of people are relying on it.

You'd think working *with* AI would be the safest place to stand. You understand it, you build with it, surely you're last in line.

It doesn't work like that, and the reason isn't ironic — it's structural.

Exposure is set by the **shape** of the work, not the subject. Does it take structured inputs? Produce code or text? Need no physical presence? Carry no licensed accountability?

Data analysis answers yes to all four. That's precisely what makes it a field where AI is useful — and precisely what makes it automatable.

A data analyst isn't protected by understanding AI, any more than a copywriter is protected by understanding language.

The scores

ROLE	SCORE
ML / AI engineer	6.0
Senior data scientist	6.5
Data engineer	7.0
Analytics engineer / BI	7.6
Entry data analyst	8.2

Two things to notice. Nothing scores below 6.0 — there's no genuinely safe corner here. And every single track moved at least a full point in three years, which no other career in our whole index does.

What separates 6.0 from 8.2 is the same thing everywhere: judgment and accountability. Building a system that makes decisions at scale has failure modes nobody has seen. Producing a requested dashboard doesn't.

Now the part that argues against us

We'd rather show you this than have you find it later and wonder what else we left out.

- **BLS projects 33.5% growth** — from 245,900 jobs in 2024 to 328,300 by 2034, roughly 23,400 openings a year
- **The World Economic Forum** ranked Big Data Specialists the single fastest-growing job in percentage terms through 2030
- **McKinsey** found 88% of organisations now use AI somewhere, and named data engineers and scientists the most in-demand AI hires, with supply lagging
- BLS median pay: **$112,590**
- And entry-level analyst salaries are reported around $90,000 — up roughly $20,000 in a year

That last one matters most. If entry-level data work were being automated away, entry-level salaries wouldn't be climbing. Prices don't usually rise when demand collapses.

So why do we hold the score?

Because our framework measures how automatable the *tasks* are, not how many people employers want. Those are different things, and here they've come apart more than anywhere else we look.

What we think is actually happening: the job title is surviving while its contents change completely. Machine learning now appears in 77% of data scientist postings. Programming expectations have migrated down from ML engineers to analysts. What made you a data scientist in 2020 makes you an analyst now.

Demand for people who can work with data is growing fast. Demand for people who can only query, clean and chart is not.

You might weigh that evidence differently from us. That's a reasonable thing to do, and it's why we've shown you both sides rather than just the number.

What actually holds up

Deciding what question is worth asking. Analysis is cheap now. Knowing what to analyse isn't. This is the single most protective skill in the field.

Knowing the data is wrong before the model does. That's domain knowledge applied to numbers, and it's the most common source of expensive mistakes.

Owning a recommendation — being the person who answers when a decision based on your analysis costs money.

Building rather than running. ML engineering and data infrastructure score better because building a system is genuinely different work from using one.

What we won't soften

Employers want two to four years' experience, which is the classic problem of needing experience to get experience.

The skill bar keeps rising. What qualified you as a data scientist in 2020 qualifies you as an analyst now — and that migration is still happening.

The job titles are a mess. "Data analyst," "data scientist," "analytics engineer," "ML engineer" describe overlapping work with enormous pay gaps, and you often can't tell which is which from the posting.

And there's no floor. No licence, no physical requirement. Every bit of protection you get, you earn.

Where to actually aim

Statistics or maths as the degree, with data skills on top. Knowing *why* a method applies survives the automation of running it. Knowing which button to press doesn't.

ML engineering and data infrastructure — the building end, which scores best.

Or data plus a domain — health, finance, energy, biology. The domain is what lets you know a result is wrong, and those employers pay a premium for it.

What doesn't work: a course that teaches Python, SQL and dashboarding aimed at an analyst job. That's training for the most automated work in the field.

Does this actually sound like you?

It suits people who:

- Are curious about **the question**, not the technique — best predictor of who moves up
- Can hold **statistical rigour and commercial context** at once
- Are comfortable saying "**the data can't tell us that**"
- Can **explain things to people who don't want the detail** — that's most of the senior job
- Don't mind that **the tooling changes constantly**

Harder if you want defined, repeatable tasks — that's the automating layer — or if you picked it because it seemed like the safe technical option. That reasoning is exactly what this document is about.

What actually matters when picking a course

- **How much is statistics and maths, versus tooling?** The single best question. Stats is the durable part.
- **Do you build systems, or just use them?**

- **Real, messy data or clean teaching sets?** The gap between those two is most of the actual job.
- **Internship placement and conversion** — given employers want experience, this is how you get around the paradox.
- **What job titles do graduates have at six months?** "Analyst" and "ML engineer" are very different outcomes.

Two things worth doing

Work with real, messy data on a real question. A competition, a project, scraping something you care about. Teaching datasets are clean; real ones aren't, and knowing what to do when the data is a mess is most of the job.

Then find someone two or three years into a data role and ask what they actually do — and what their job title means at their company. The answer is often not what you'd guess.

One last thing

Read the BLS page yourself. It's free, it's linked below, and it says something considerably more optimistic than our score does.

We'd rather you looked at both and formed your own view than took our number on trust. We think we're right that the entry tier is exposed. We might be wrong about how much that matters when demand is growing this fast.

What we're confident about: the protected end of this field is real, and it's further up than the course brochures suggest.

If you want to go further — three things worth twenty minutes

- **Data Scientists — Occupational Outlook Handbook** — read this one first. It's the government's own projection, it's free, and it argues against us.
- **Data scientist job outlook 2026** — based on 1,000+ real job postings. What employers actually ask for, and how fast the requirements are moving.

- **Data analyst job outlook 2026** — the same for the analyst tier, including why specialists are beating generalists.

There's a longer version behind this — full scores by role, how they were worked out, the counterargument in detail, what to ask a program, and where we might be wrong. Your parents have it.